Generating functions

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Sequences and formal power series

The ordinary generating function (OGF) of $(a_n)_{n\geq 0}$ is $A(x) = \sum_{n\geq 0} a_n x^n$. With formal indeterminate x, neither numerical substitution nor convergence is required; coefficient comparison remains valid. Numerical real or complex evaluation additionally requires analytic convergence. Finite sequences give polynomials.

For a_n=1, $A(x) = \sum_{n\geq 0} x^n = (1-x)^{-1}$, meaning all coefficients agree. Formal addition and multiplication are defined algebraically. Division requires an invertible constant coefficient in the denominator, equivalent to a nonzero constant over a field.

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Which formal operations are legitimate?

A series is a coefficient sequence. Equality means equality at every degree. Thus $1/(1-x) = 1 + x + x^2 + \cdots$ first means $(1-x)(1+x+x^2+\cdots) = 1$, rather than a numerical limit. Convolution uses only finitely many terms at each fixed degree.

For $A(x) = \sum a_n x^n$ with $a_0 \neq 0$, its inverse $B(x) = \sum b_n x^n$ is determined by

$$b_0 = 1/a_0, \qquad a_0 b_n + \cdots + a_n b_0 = 0, \qquad b_n = -\frac{1}{a_0} \sum_{k=1}^{n} a_k b_{n-k} \quad (n \geq 1).$$

Only earlier b coefficients occur, and the fixed nonzero denominator makes this recursion valid for every n.

An infinite product is legitimate when only finitely many factors affect each fixed coefficient. In $\prod_{k \geq 1} (1 - x^k)^{-1}$, factors k>n equal one through degree n, satisfying this condition.

Which formal operations are legitimate? (continued)

Formal power series

Compare coefficients

Add, multiply, differentiate formally

1/(1−x) is an inverse, not a limit

Operations follow formal algebra

Analytic functions



Check |x|<R before evaluation

Coefficient identities are formal; numerical values need analytic conditions

Numerically the geometric series converges for |x|<1. That condition is additional to, rather than a premise of, the formal coefficient identity.

Operations and convolution

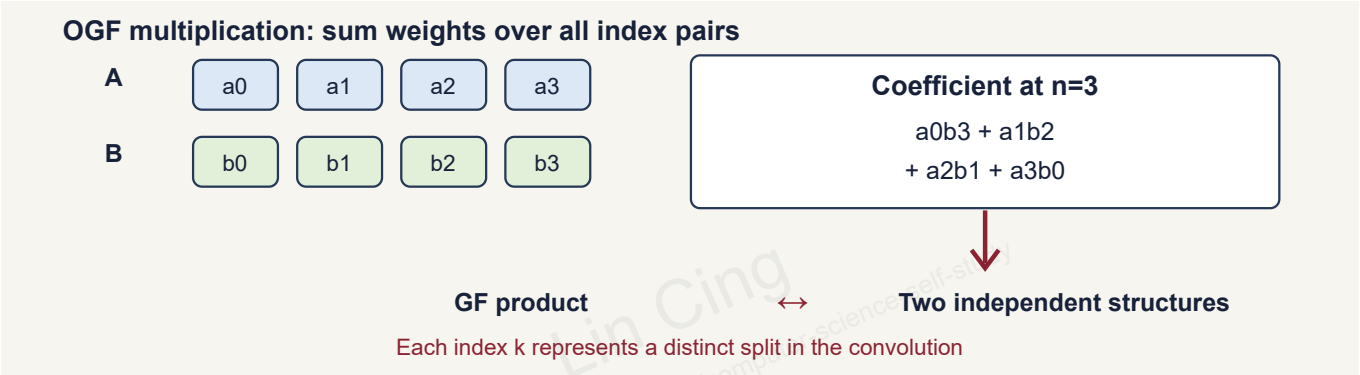
For A and B with coefficients a_n,b_n,

$$A(x)B(x) = \sum_{n\geq 0}\left(\sum_{k=0}^{n} a_k b_{n-k}\right)x^n.$$

Multiplying by x^j shifts indices upward by j with j leading zeros; scalar multiplication scales coefficients. Convolution models a split into two independent smaller objects, summed over every possible split size.

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Operations and convolution (continued)



If objects have additive weight n and count a_n, define $A(x) = \sum a_n x^n$. Coefficients recover the counts once a counting rule or bijection is established. The product rule requires a unique decomposition into an ordered pair (u,v), with additive weights. Nonunique decompositions repeat objects. A partition factor $1/(1-x^k)$ records multiplicity of a fixed part size, so 3+1 and 1+3 do not differ; ordered compositions require a different model.

Product models and coin combinations

An allowed multiplicity set S gives factor $\sum_{r \in S} x^r$. Independent choices across types multiply; the degree-n coefficient counts total weight n. Coefficients are nonnegative integers counting permitted choices.

Unlimited coins of values 1,2,3 give $1/((1-x)(1-x^2)(1-x^3))$. Identical coins record quantities, not order. Ordered coin sequences instead use $D(x)=x+x^2+x^3$; length k gives $D(x)^k$ and all lengths, including empty, give $1/(1-x-x^2-x^3)$. At total three, unordered choices are 1+1+1,1+2,3; ordered sequences also include 2+1, giving four. The change is order, not invented labels on identical coins.

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Bounded choices and negative binomial series

Multiplicity at most m gives $1+x+\cdots+x^m=(1-x^{m+1})/(1-x)$; exactly m gives x^m; a minimum a extracts x^a. Three types each allowed 0,1,2 have count $[x^n](1+x+x^2)^3$. At n=5 only permutations of (2,2,1) contribute, giving three; unrestricted stars and bars $\binom{7}{2}=21$ ignores the upper bound.

Four unlimited types give $(1-x)^{-4}=\sum_{n\geq 0}\binom{n+3}{3}x^n$. Generally for positive integer r,

$$(1-x)^{-r}=\sum_{n\geq 0}\binom{n+r-1}{n}x^n.$$

The coefficient counts $y_1+\ldots+y_r=n$ with nonnegative y_i. Arrange n stars and r−1 bars in $\binom{n+r-1}{r-1}=\binom{n+r-1}{n}$ ways, bijectively encoding all allocations. Requiring each type positive first places one of each, giving factor $x^r/(1-x)^r$ and coefficient $\binom{n-1}{r-1}$ for $n\geq r$. Unlimited and bounded multiplicities require different factors and sample spaces.

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Partial fractions and coefficient extraction

For example,

$$\frac{1}{(1-x)(1-2x)} = \frac{2}{1-2x} - \frac{1}{1-x} = \sum_{n\geq 0}(2^{n+1}-1)x^n,$$

with first coefficients 1,3,7,15. Preserve proper numerator degrees, all repeated-factor powers, and irreducible quadratic terms. A finite truncation cannot determine omitted coefficients.

For P(x)/Q(x), first divide polynomials if deg P≥deg Q. Factor Q over the reals into linear and irreducible quadratic factors. A repeated factor $(1-ax)^m$ requires every term $C_1/(1-ax)+\ldots+C_m/(1-ax)^m$. An irreducible quadratic normally requires a linear numerator. Extract using

$$[x^n](1-ax)^{-k} = a^n\binom{n+k-1}{k-1}.$$

An irreducible real quadratic may still simplify:

$$\frac{1}{1+x+x^2} = \frac{1-x}{1-x^3}, \qquad (c_0, c_1, c_2, \ldots) = (1, -1, 0, 1, -1, 0, \ldots).$$

Partial fractions and coefficient extraction (continued)

The numerator 1−x is essential. Expanding $1-(x+x^2)+(x+x^2)^2-\ldots$ also works, but every term contributing to a fixed degree must be retained. Rational partial-fraction constants can combine into integer coefficients when the model requires integers.

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Formal differentiation and products

Differentiate $(1-x)^{-1} = \sum_{n\geq 0} x^n$:

$$(1-x)^{-2} = \sum_{n\geq 1} n x^{n-1} = \sum_{m\geq 0} (m+1) x^m.$$

Repeating r−1 times yields $(1-x)^{-r} = \sum_{m\geq 0} \binom{m+r-1}{m} x^m$. Formal differentiation only transforms coefficients and requires careful indices, not convergence. Analytic series and derivatives share a convergence radius but may differ at endpoints. Numerical comparisons require the convergence domain; finite polynomial truncations do not supply the missing coefficients. Low-degree expansion and stars-and-bars counts check the identity independently.

Probability-generating coefficients

For nonnegative integer-valued X, define $P_X(z) = \sum_{k\geq 0} \Pr(X=k) z^k$. Coefficients are point probabilities, not densities. Nonnegative coefficients summing to one give P_X(1)=1.

For equal probabilities on 2,4,6,

$$P_X(z) = (z^2 + z^4 + z^6)/3, \qquad E[X] = P_X'(1) = (2+4+6)/3 = 4.$$

Also

$$P_X''(1) = (2 + 12 + 30)/3 = 44/3 = E[X(X-1)],$$
$$E[X^2] = 44/3 + 4 = 56/3, \qquad \operatorname{Var}(X) = 56/3 - 16 = 8/3.$$

For infinite support, such as geometric distributions, differentiation and evaluation at one require justified limits or convergence and need not give a finite derivative. Formal coefficient identities alone do not establish finite moments; check support, normalization, and convergence.

Integer partitions

The count p(n) of unordered positive-part sums has OGF

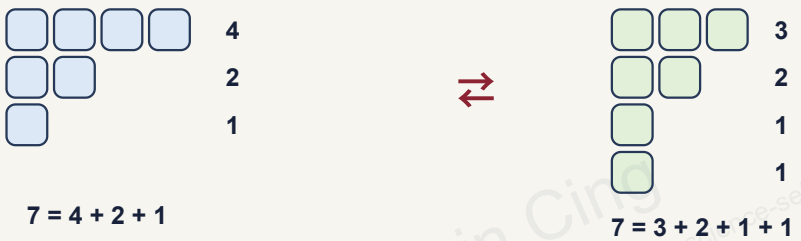
$$\prod_{k\geq 1}(1-x^k)^{-1}.$$

Each factor selects a multiplicity of part size k. Thus 3+1=1+3 as partitions. Ordered compositions, including empty, instead give $1/(1-(x+x^2+\cdots)) = (1-x)/(1-2x)$; nonempty compositions give x/(1−2x).

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Integer partitions (continued)

Ferrers transposition: a bijection with conjugate partitions



Same number of cells

Largest part 4 ⇄ 4 parts

3 parts ⇄ largest part 3

Equal family sizes

At any degree n, only factors k≤n matter, making the infinite product formally valid. Fixed part-size factors remove order by retaining only multiplicity; their sequence in the product is fixed rather than permuting parts.

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Bounds on the largest part

Largest part at most m gives $\prod_{k=1}^{m}(1-x^k)^{-1}$. At m=2, choose j twos and n−2j ones, for j=0,…,⌊n/2⌋, yielding ⌊n/2⌋+1 partitions.

Ferrers transposition exchanges largest part and number of parts. For $\lambda_1 \geq \cdots \geq \lambda_\ell > 0$, define $\lambda'_j = |\{i : \lambda_i \geq j\}|$. New rows are old column lengths, preserving total cells; $\lambda'_1 = \ell$ and the new number of parts is λ_1. A cell (i,j) exists iff $j \leq \lambda_i$ iff $i \leq \lambda'_j$; transposing again restores the original cells. This proves a reversible bijection, so at most m parts and largest part at most m are equinumerous. Distinct parts replace factors by $1+x^k$; at least one of a given size uses $x^k/(1-x^k)$.

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Odd parts and distinct parts

The product identity

$$\prod_{k\geq 1}(1+x^k) = \prod_{k\geq 1}\frac{1-x^{2k}}{1-x^k} = \prod_{k\geq 1}(1-x^{2k-1})^{-1}$$

comes from canceling even factors, proving equality of distinct-part and odd-part partition counts.

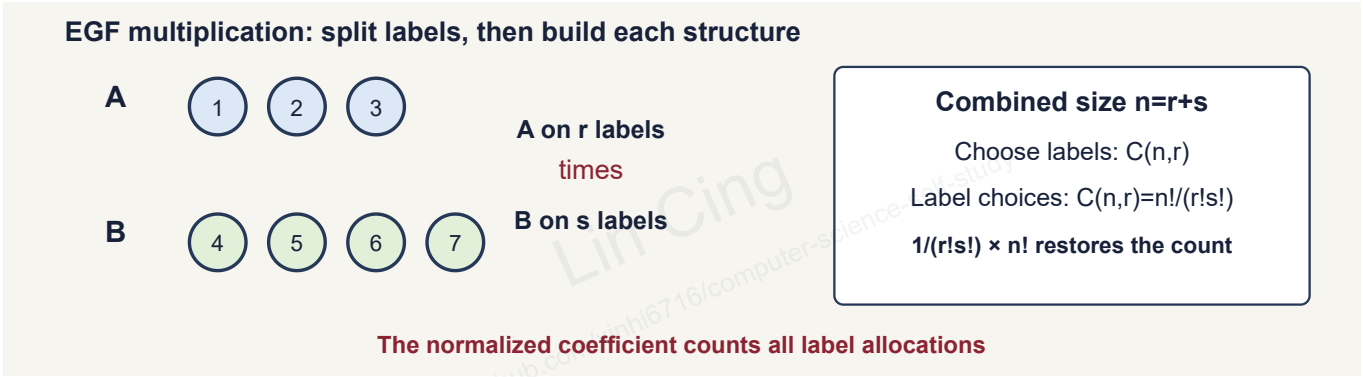
For an explicit bijection, write each odd part's multiplicity in binary. A nonzero bit 2^j creates one part of size 2^j times that odd part. Unique power-of-two-times-odd decomposition ensures distinct parts and a unique inverse.

For 26=5+5+5+3+3+1+1+1+1+1, multiplicities 3=1+2,2=2,5=1+4 give {5,10},{6},{1,4}, hence 10+6+5+4+1. Reverse 10 to two fives, 6 to two threes, 5 to one five, 4 to four ones, 1 to one one. Odd families cannot collide because their odd factors are unique.

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Exponential generating functions

For labeled objects define $\widehat{A}(x) = \sum_{n\geq 0} a_n x^n/n!$. Splitting n=r+s labels into two groups gives $\binom{n}{r} = n!/(r!s!)$, which is exactly the normalization supplied by an EGF product.



If a_r counts A-structures on any r labels and b_s B-structures on s disjoint labels, first choosing the A-label subset then both structures gives $\sum_{r=0}^{n} \binom{n}{r} a_r b_{n-r}$. Algebraically,

Exponential generating functions (continued)

$$[x^n](\widehat{A}\widehat{B}) = \sum_{r=0}^{n} \frac{a_r b_{n-r}}{r!(n-r)!} = \frac{1}{n!} \sum_{r=0}^{n} \binom{n}{r} a_r b_{n-r}.$$

Multiplication by n! restores the count. The normalization encodes label splitting rather than an arbitrary conversion from an OGF.

Partitions into unlabeled nonempty blocks give $\exp(e^x - 1) = \sum B_n x^n/n!$. Initial Bell numbers are 1,1,2,5. Two labels either share one block or separate; three labels give one full block, three singleton-plus-pair choices, or three singleton blocks, totaling 1+3+1=5.

Why SET components must be nonempty

Let $\hat{A}(0)=0$ count nonempty labeled components. Fixed k distinguishable slots have EGF $\hat{A}(x)^k$. Disjoint nonempty label sets make the components distinct objects, so every unordered component set has exactly k! slot orders. Therefore

$$\widehat{C}_k(x) = \widehat{A}(x)^k/k!, \qquad \widehat{C}(x) = \sum_{k\geq 0} \widehat{A}(x)^k/k! = \exp(\widehat{A}(x)).$$

At degree n only $k \leq n$ contributes. Empty components would allow arbitrarily many empty blocks and invalidate this interpretation and finite layering.

One nonempty set component has EGF e^x-1. For n=4,k=2, ordered sizes (1,3),(2,2),(3,1) contribute 4,6,4. Division by 2! gives seven, so $4![x^4](e^x-1)^2/2! = 7$. The four-label counts for k=1,2,3,4 are 1,7,6,1, totaling B_4=15.

Choose k companions of a new label in its block; partition all remaining labels. These reversible choices give

Why SET components must be nonempty (continued)

$$B_{n+1} = \sum_{k=0}^{n} \binom{n}{k} B_{n-k},$$

also obtained from $B'(x)=e^xB(x)$ by the EGF product rule.

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Repeated letters and parity restrictions

With positions as labels and letter i allowed multiplicities J_i,

$$\#\text{words} = n![x^n] \prod_i \left(\sum_{j \in J_i} x^j / j! \right).$$

For fixed multiplicities summing to n, the labeled product assigns positions in $n!/\prod j_i!$ ways, precisely the repeated-letter arrangement count. OGF multiplicity selection would ignore these orders.

With at most two A's, two B's, one C, length three gives

$$3!\,[x^3](1 + x + x^2/2)^2(1 + x) = 6(1 + 2) = 18.$$

Multiplicities (2,1,0),(1,2,0),(2,0,1),(0,2,1) each give three words; (1,1,1) gives six. Length four gives 4! (1/4+1)=30. Identical letters receive no artificial inventory labels.

Even A-count, with B,C unrestricted, gives

$$n![x^n]\frac{e^x + e^{-x}}{2} e^{2x} = (3^n + 1)/2.$$

Repeated letters and parity restrictions (continued)

The even filter cancels odd powers. At n=4, A-counts 0,2,4 contribute 16,24,1, totaling 41. Exponentials here are formal series, without numerical substitution.

The summation operator

For $A(x) = \sum a_n x^n$ and $s_n = \sum_{k=0}^{n} a_k$, convolution with all ones gives

$$A(x)(1 + x + x^2 + \cdots) = \sum_{n \geq 0} \left(\sum_{k=0}^{n} a_k \right) x^n = S(x) = A(x)/(1 - x).$$

Thus 1/(1−x) is the sum operator; repeating it gives higher prefix sums. Weighted sums or a starting index of one require separate shifts. No analytic convergence is needed.

For a_k=2k+1, prefix sums are $(n+1)^2$, hence $A(x)/(1-x) = \sum_{n \geq 0} (n+1)^2 x^n$. Combining summation with x(d/dx) also derives power-sum identities.

For nonnegative u_1, u_2, u_3 with $u_1+2u_2+3u_3 \leq n$, add the unique nonnegative slack u_4 so $u_1+2u_2+3u_3+u_4=n$. The four factors give

$$\sum_{n \geq 0} A_n x^n = \frac{1}{(1-x)^2(1-x^2)(1-x^3)} = 1 + 2x + 4x^2 + 7x^3 + 11x^4 + 16x^5 + 23x^6 + 31x^7 + \cdots .$$

Thus A_7=31, agreeing with direct enumeration of the triples. The extra factor follows from slack, preserving the inequality's exact meaning.

Fibonacci OGF and both closed-form derivations

For F_0=0,F_1=1 and $F_n = F_{n-1} + F_{n-2}$, sum shifted recurrences:

$$F(x) - x = xF(x) + x^2 F(x), \qquad F(x) = x/(1 - x - x^2).$$

The denominator constant is one, so formal division is valid. The first coefficients are 0,1,1,2,3,5. Comparing $(1 - x - x^2)F(x) = x$ gives constant F_0=0, linear $F_1 - F_0$=1, and for n≥2 exactly $F_n - F_{n-1} - F_{n-2}$=0, verifying every coefficient and initial shift.

Let $\varphi = (1 + \sqrt{5})/2, \psi = (1 - \sqrt{5})/2$, the roots of $r^2 - r - 1 = 0$. Then $1 - x - x^2 = (1 - \varphi x)(1 - \psi x)$, $\varphi - \psi = \sqrt{5}$, and

$$F(x) = \frac{1}{\sqrt{5}} \left(\frac{1}{1 - \varphi x} - \frac{1}{1 - \psi x} \right).$$

Its numerator is $((1 - \psi x) - (1 - \varphi x))/\sqrt{5} = x$. Extracting coefficients gives $F_n = (\varphi^n - \psi^n)/\sqrt{5}$.

Independently, the characteristic-root expression $F_n = A\varphi^n + B\psi^n$ has initial equations A+B=0 and $A\varphi + B\psi = 1$, giving $A = 1/\sqrt{5}, B = -1/\sqrt{5}$ and the same formula. Both routes need both initial values; roots alone do not specify the sequence. Formal identities hold for every n; analytic substitution requires $|x| < 1/\varphi$. Chapter 10 applies the method to general recurrences.

Boundaries and checks

Keep coefficient indices and polynomial degrees aligned, especially at n=0. Distinguish formal identities from analytic evaluation. Multiplicity limits, ordered sequences, and labels each change factors. Check extracted coefficients using a recurrence or small cases. The exercise and solution PDFs provide further derivations. A valid formula follows from a bijection or coefficient identity that can be expanded, rather than resemblance alone.

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References and attribution

Ralph P. Grimaldi. Discrete and Combinatorial Mathematics: An Applied Introduction, 5th Edition, 2004.

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